MIL-D-81694 (AS)
2 February 1972

MILITARY SPECIFICATION

DISPENSING SET, COUNTERMEASURES AND CHAFF AN/ALE-39

This specification has been approved by the Naval Air Systems Command, Department of the Navy

1. (U) SCOPE

1.1 (C) Scope.- The AN/ALE-39 dispensing set covered by this specification shall provide the capability of dispensing multiple ECM decoys from Naval aircraft. The AN/ALE-39 shall provide the flexibility to selectively dispense three types of payloads: chaff, flares, and jammers.

1.2 (U) Classification.- The AN/ALE-39 covered by this specification shall consist of the following items:

	Quantity	Applicable Paragraph
Programmer, Dispensing Set MX-9254/ ALE-39	(1)	3.5.2
Switch, Sequencer SA-1874/ALE-39	(2)	3.5.3

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by authority of Naval Air Sys Com on 10 Jan 1975
NAVAIRSYSCOM ltr 5262/3:PY/ of 21 Jan 75
Signed: P. Yurcisin

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1.3 (U) Associated equipment.- This equipment shall operate with the associated equipment listed in 6.7.

2. (U) APPLICABLE DOCUMENTS

2.1 (U) General - The following documents of the issue in effect on the date of invitation for bids form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-W-5088	Wiring; Aircraft, Installation of
MIL-E-5400	Electronic Equipment, Aircraft, General Specification for
MIL-[illegible]22	Testing, Environmental, Aircraft Electronic Equipment
MIL-C-6781	Control Panel; Aircraft Equipment, Rack or Console Mounted
MIL-E-17555	Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of
MIL-T-18303	Test Procedures; Preproduction, Acceptance, and Life for Aircraft Electronic Equipment, Format for
MIL-N-18307	Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment
MIL-C-81774	Control Panel, Aircraft, General Requirements for

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Naval Air Systems Command

AR-5	Microelectronic Devices Used in Avionic Equipment, Procedures for Selection and Approval of
AR-10	Maintainability of Avionics Requirement and Systems, General Requirements for
AR-34	Failure Classification for Reliability Testing, General Requirement for

STANDARDS

Military

MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipment
MIL-STD-704	Electric Power, Aircraft Characteristics and Utilization of
MIL-STD-781	Reliability Tests, Exponential Distribution
MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of
MIL-STD-810	Environmental Test Methods (Acoustic Stimulation, Level B only)

2.1.1 (U) Availability of documents.- When requesting specification, standards, drawings, and publications, refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Publication and Forms Center, Code 105, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

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3. (U) REQUIREMENTS

3.1 (U) Preproduction.- This specification makes provision for preproduction testing.

3.2 (U) Parts and materials.- In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing the following shall govern:

(1) Microelectronic items shall conform to requirements specified herein.

(2) Other parts and materials requirements shall conform to specification MIL-E-5400.

(3) Nonrepairable subassemblies shall be used in accordance with requirement AR-10 as outlined in specification MIL-E-5400.

(4) When previously produced models of this equipment did not use nonrepairable subassemblies, the design shall not be changed to employ nonrepairable assemblies without the approval of the procuring activity.

3.2.1 (U) Nonstandard parts and materials approval.- Approval for the use of nonstandard parts and materials other than microelectric devices shall be obtained as outlined in specification MIL-E-5400. Microelectronic devices (MIC) shall be approved as outlined in AR-5.

3.2.2 (U) Microelectronic modular assemblies (MICRA).- When used, Microelectronic modular assemblies shall meet the requirements of AR-5.

3.2.3 (U) Modules, maintenance.- The electronic portions the equipment shall be divided into maintenance modules (see AR-10). Maintenance modules shall normally be considered repairable.

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3.3 (U) Design and construction.- The equipment shall conform to all the applicable requirements of specification MIL-E-5400 for design, construction, and workmanship, except as otherwise specified herein.

3.3.1 (U) Weights.- The weight of the Programmer Unit shall not exceed 2.5 pounds. The weight of each Sequence Switch shall not exceed 3.5 pounds.

3.3.2 (U) Reliability.- The contractor shall conduct a reliability program using MIL-STD-785 as a guide. On a reorder from a supplier who has previously produced the equipment, the program previously used may be continued unless otherwise indicated in the contract or order.

3.3.2.1 (U) Operational stability.- The equipment shall operate with satisfactory performance, continuously, or intermittently for a period of at least 500 hours with exception of the clock. The clock will have at least a 50 hour operational stability at which only an adjustment may be required.

3.3.2.2 (U) Operating life.- The equipment shall have total operating life of 5000 hours with reasonable servicing and replacement of parts. Parts requiring scheduled replacement shall be specified by the contractor.

3.3.2.3 (U) Reliability in Mean Time Between Failures (MTBF).- The equipment shall have 500 hours of mean (operating) time between failures when tested and accepted as outlined under the requirements of 4.4.3.

3.3.3 (U) Cabling and connections.-

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3.3.3.1 (U) Cables and connectors.- The equipment shall provide for the use of cables and connectors in accordance with specification MIL-E-5400.

3.3.3.2 (U) Interconnection cabling.- The equipment shall be capable of satisfactory operation using external wiring in accordance with the applicable requirements of specification MIL-W-5088. The external wiring shall be unshielded, except that a minimum number of the individual wires may be shielded when demonstrated as necessary to meet interference control requirements and provided the assembly of the cable to its plugs may be easily accomplished. External cables and that portion of the connectors attached to the cables shall not be supplied as part of the equipment.

3.3.4 (U) Control panels.- All rack or console mounted control panels shall conform to the applicable requirements of specification MIL-C-6781. The configuration of all panels must be approved by the procuring activity prior to preproduction testing.

3.3.5 (U) Interchangeability.-

3.3.5. (U) AN/ALE-39.- All parts and components for any one dispenser system shall be physically and functionally interchangeable with all corresponding parts and components produced under contract in accordance with specification MIL-E-5400. The two sequencer switches in the dispenser system shall be physically and functionally interchangeable with each other.

3.3.6 (U) Interference control.- The equipment shall be defined as being within the category of "Computers", and shall operate within the limits specified by the tests listed in Table IV, (Airborne Data Processing and Recording) of MIL-STD-461A, Specific requirements of the equipment either augmenting or deleting those listed herein shall be enumerated in the purchase description by the procuring activity.

3.3.6.1 (U) Applicable tests.- The tests applicable to the equipment are as follows: CE01, CE02, CE03, CE04, CE05, CS01, CS02, CS06, RE02, RS01, RS02, RS03.

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3.3.7 (U) Provisions for maintainability.- The maintainability program, construction and packaging, provisions for test points, and other maintainability parameters shall be as specified in specification AR-10.

3.3.8 (U) Nomenclature and nameplates.- Nomenclature assignment and nameplate approval for equipment identification shall be in accordance with specification MIL-N-18307.

3.3.9 (U) Standard conditions. - The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

Temperature:	Room ambient (25°C ± 5°C)
Altitude:	Normal ground
Vibration:	None
Humidity:	Room ambient up to 90 percent relative humidity
Input power voltage:	27.5 ± 0.5 V DC

3.3.10 (U) Service conditions.- The equipment shall operate satisfactorily under any of the environmental service conditions or reasonable combination of these conditions as specified in specification MIL-E-5400 for Class 4 equipment, except as modified herein.

3.3.10.1 (U) Vibration.- The equipment shall operate satisfactorily when subjected to the vibration requirements of Curve I of specification MIL-E-5400.

3.3.10.2 (U) Temperature. -

3.3.10.2.1 (U) The programmer shall operate continuously at a maximum of +45°C. For the purpose of MIL-E-5400, short-time or intermittent temperature extremes do not apply.

3.3.10.2.2 (U) The sequencer switch shall operate short-time at +150°C; the intermittent temperature extreme does not apply.

3.3.10.3 (U) Altitude. -

3.3.10.3.1 (U) The programmer and sequencer switch shall operate to a maximum altitude of 70,000 feet.

3.3.10.4 (U) Temperature - Altitude -

3.3.10.4.1 (U) The programmer and sequencer switch shall operate satisfactorily under the conditions imposed for Class 3 equipment, Curve A, Figure 3 (sheet 3 of 4), MIL-E-5400, except the altitude shall be 70,000 feet maximum.

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3.3.10.5 (U) Acoustic stimulation.-

3.3.10.5.1 (U) The sequencer switch shall be insensitive to acoustic stimulation at a level of category B described in MIL-STD-810.

3.3.11 (U) Warm-up time.- The time required for the equipment to warm-up prior to operation shall be kept to a minimum and shall not exceed 0.5 seconds under standard conditions and 1.0 second at extreme service conditions.

3.3.12 (U) Input electrical power.-

3.3.12.1 (U) Operating power.- The AN/ALE-39 dispenser system shall meet all applicable requirements of MIL-STD-704 and shall give specified performance when energized from power sources having characteristics and limits as defined in MIL-STD-704. The power source required shall not exceed the specified amount of 27.5 ± 1.5 V DC, Category B, 20 amps.

3.3.12.1 (U) Equipment protection.- The AN/ALE-39 cockpit control unit, programmer, and sequencer switches shall be self-protected against damage by excessive voltage surges as determined by MIL-STD-704, Category B.

3.3.12.2 (U) Lighting power.- Input power for lighting of the programmer shall require not more than 10 watts at 5.0 volts, AC, 400 Hz, or 14/28 volts, AC 400 Hz. No lighting power is required for the dispenser or sequencer switch.

3.3.12.3 (U) Degraded performance.- Degraded performance will be permitted for voltage transients not exceeding 0.5 second during normal electric system operation. Operation shall return to normal with no resulting damage to the equipment.

3.3.12.4 (U) Standby power.- The equipment shall provide for a standby position whereby rated voltage shall be applied to all circuits. The standby power shall not exceed 900 mw.

3.3.13 (U) Cooling.- No external cooling air is required by the dispenser system.

3.3.13.1 (U) Wiring harness.- The wiring harness for interconnecting the components of the dispenser system is located in the aircraft and is depicted in Figure 1.

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3.4 (U) Performance.- Unless otherwise specified, values set forth to establish the requirements for satisfactory performance apply to performance under both standard and extreme service conditions. When reduced performance under the extreme conditions is acceptable, tolerances or values setting forth acceptable variations from the performance under the standard conditions will be specified.

3.4.1 (U) Accuracy.- Accuracy shall be ±5 percent.

3.4.2 (U) Operation.- The equipment shall provide for selectively dispensing three types of expendable payloads automatically or manually.

3.5 (U) Detail requirements.-

3.5.1 (U) Control unit.- The individual control units shall be covered under separate specification. Operating characteristics are specified in Appendix A.

3.5.2 (U) Programmer.-

3.5.2.1 (C) Function.- The AN/ALE-39 programmer shall generate the control signals required for dispensing expendable payloads of chaff, flares, and jammers in either automatic or manual modes. Three programs shall be provided, one for each of the three types of payloads. The programs shall each be capable of independently firing the specified number of respective payloads from either or both dispensers loaded with 10, 20, 30, 40, 50, or 60 payloads. The programmer shall be capable of independently and simultaneously (within 0.075 seconds) dispensing different types of payloads by their respective programs.

The programmer shall operate with the control unit described in Appendix A through an aircraft harness described in Figure 1. The harness contains 7 wires between control and programmer units of which 3 wires are allocated for command signals, 3 wires for payload counter signals, and 1 wire for the multiple flare control. In aircraft which do not have the 7 wire harness available, reduced control unit function will result unless additional information transmission capability is provided GFE.

3.5.2.2 (U) Form factor.- The programmer shall be packaged in a 3 5/16 by 3 3/4 by 5 inch case with an edge-lighted panel, and shall be physically interchangeable with the AN/ALE-29A programmer unit mounting provisions. Reference Figure 2.

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3.5.2.2.1 (U) Panel. - The lettering on the panel shall be 0.140 inch high in accordance with MIL-D-81774 paragraph 4.6.5.3 The panel shall b edge-lighted and the lamps shall be 5 V AC or 14/28 V AC (400 Hz) conforming to requirements of each aircraft.

3.5.2.3 (U) Weight.- The weight of the programmer shall not exceed 2.5 pounds.

3.5.2.4 (U) Contents.- The programmer shall contain switches, circuits, and components required to perform the functions of paragraph 3.5.2.1.

3.5.2.5 (U) Controls.-

3.5.2.5.1 (U) Automatic mode.- Each of the three programs shall have controls for use in automatic mode as listed below and summarized in Table I.

3.5.2.5.1. (U) Chaff controls.-

(1) (C) A control labeled "B-QTY" shall determine the number of chaff loads to be dispensed on one salvo. Provision shall be made for selection of 1, 2, 3, 4, C (continuous) or R (random) burst(s). In random position a minimum of 4 total bursts and a maximum total of 6 bursts shall be dispensed automatically with the exact number fired equi-probably determined.

(2) (C) A control labeled "B-INTV" shall determine the time between bursts in a particular salvo. Provision shall be made for the selection of 0.125, 0.25, 0.75, 1.0, or R (random) second intervals. The burst interval is defined as the time between the start of one burst to the start of the succeeding burst within the salvo and is equal to the burst cycle time. In random "B-INTV" position the time for the first three burst intervals shall automatically be 0.125 seconds with additional random quantities of 1, 2, or 3 bursts being fired at equi-probable random time intervals of either 0.25, 0.5, 0.75, 1.0, 1.5, 2.0, 3.0, or 4.0 seconds whenever the burst qantity control is set at R or C. When the burst interval is set at R and the burst quantity control is not on R or C, a single chaff payload will be dispensed.

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(C) TABLE I

PROGRAMMER PARAMETERS IN AUTOMATIC MODE
FOR AN/ALE-39 DISPENSER

Type of Payloads:	CHAFF	FLARES	JAMMER
Burst a	1 2 3 4 C R	2 3 4 6 8 10	1 2 3 4
Burst Intervals (seconds)	0.125 0.25 0.5 0.75 1.0 R	2 c 4 6 8 10	1 to 299 d
Salvos b	1 2 4 6 8 10 15	None	None
Salvo Intervals (seconds)	2 4 6 8 10	None	None

C = Continuous R = Random

a Burst - Individual ejection of one payload

b Salvo - Single sequence of burst

c A rapid fire sequence capable of being engaged from cockpit control unit shall be provided to eject all flares sequentially with a burst interval of 0.125

d Burst intervals shall be variable in 1 second increments between 1 and 299 seconds with an accuracy of ±5 percent.

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TABLE I (Continued)

NOTES:

1. (C) Typical chaff combinations of Burst Quantity and Burst Interval include the following:

   A. Burst Quantity (C) and Burst Interval (R).- First 3 payloads at 0.125 second intervals with the total remaining payloads dispensed at the pseudo-random time intervals of 0.25, 0.50, 0.75, 1.0, 1.5, 2.0, 3.0, or 4.0 seconds. Under these conditions the Salvo Quantity and Salvo Interval control functions are negated.

   B. Burst Quantity (R) and Burst Interval (R).- First 3 payloads at 0.125 second intervals followed by 1, 2, or 3 bursts at the pseudo-random time intervals of 0.25, 0.50, 0.75, 1.0, 1.5, 2.0, 3.0, or 4.0 seconds. The number of bursts and time intervals shall have equal probability of occurrence. Any additional salvos will occur at the salvo interval and will have 4 to 6 bursts but will not repeat the 3 - 0.125 second bursts.

   C. Burst Quantity (R) and Burst Interval (Fixed No.).- First 3 payloads at 0.125 second intervals followed by 1, 2, or 3 bursts at the fixed time interval. The number of bursts shall have equal probability of occurrence. Any additional salvos will occur at the salvo interval and will have 4 to 6 bursts, but will not repeat the 3 - 0.125 second bursts.

   D. Burst Quantity (Fixed No.) and Burst Interval (R).- One payload will be dispensed equivalent to a CHAFF MANUAL.

2. (C) Definition:

   A. Pseudo-ramdom time intervals are interpreted to mean that a pattern of 126 (minimum) different combinations of the eight time intervals mentioned above, will occur before the pattern repeats.

3. (C) The probability of starting at any point in the pattern of pseudo-random time intervals shall be equal to the reciprocal of the number of combinations (e.g., 1/126 maximum).

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3.5.2.5.1.1 (U) Chaff controls.- (Continued)

(3) (C) A control labeled "S-QTY" shall determine the number of salvos in a particular firing sequence. The control shall provide for selection of 1, 2, 4, 6, 8, 10, or 15 salvos.

(4) (C) A control labeled "S-INTV" shall determine the time between salvos. The control shall provide for selection of 2, 4, 6, 8, or 10 second intervals.

3.5.2.5.1.2 (U) Flare controls.-

(1) (C) A control labeled "QTY" shall determine the number of flares to be dispensed. The control shall provide for selection of dispensing 2, 3, 4, 6, 8, or 10 flares.

(2) (C) A control labeled "INTV" shall determine the time between bursts. The control shall provide for selection of 2, 4, 6, 8, or 10 seconds time intervals.

3.5.2.5.1.3 (U) Jammer controls.

(1) (C) A control labeled "QTY" shall determine the number of jammers to be dispensed. The control shall provide for selection of dispensing 1, 2, 3. or 4 jammers.

(2) (C) A control labeled "INTV" shall determine the time interval between jammer bursts. The controls shall provide for selection of 1 to 299 second intervals in 1 second steps. The timeing accuracy of the controls shall be ±5 percent.

3.5.2.5.2 (C) Manual mode.- The programmer shall provide for immediate (within 0.125 second) dispensing a single payload each time the manual mode switch on the cockpit mode control for the desired type of payload is initiated. The programmer shall also provide for the pilot's chaff manual control input. The manual mode of dispensing shall "overlay" the automatic mode and shall not cause interruption of the automatic program such that it must be physically reinitiated.

3.5.2.5.3 (U) Dispenser controls.- Four controls labeled "LOAD" L-10, L-20, R-10, R-20 shall be provided to indicate the type of dispensable payload loaded in the corresponding section of the dispenser.

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3.5.2.5.3.1 (U) Payload Priority.- The priority for dispensing like payloads shall be L-10, L-20, R-10, R-20. Routing and transfer of the control signals to the sequencer switch shall be automatically determined and accomplished by the programmer.

3.5.2.5.4 (U) Flare multiple.- The programmer shall provide the capability of dispensing 2, 3, or 4 flares in parallel, as determined by the dispenser controls (ref. 30.1.5.2.3) and dispenser loading pattern in either the automatic or manual modes.

3.5.2.5.5 (C) Flare-Rapid Fire sequence.- The programmer shall provide for parallel rapid firing of all flares in a sequential firing fashion with a maximum time of 0.125 seconds between firings. The total time required to fire all flares will be determined by the flares remaining in the dispenser.

3.5.2.5.6 (U) Reset control.- A reset function shall be actuated by guarded toggle switch on the programmer panel to return the programmer and sequence switches to their initial conditions.

3.5.2.5.7 (U) Power loss.- An interruption of power will stop the dispensing action if in progress, and will reset the programmer to the starting position of the dispensing program being used. The sequence switch will retain the position of the next available payload (3.5.3.1.2). These functions shall remain as they exist in the current AN/ALE-29A dispenser system.

3.5.2.6 (U) Electrical connections.- Connections to the control unit and sequencer switches shall be provided through a single connector (J1) type MS3112E-14-19PX.

3.5.3 Sequencer switch.-

3.5.3.1 (U) Function.- The sequencer switch shall meet the service conditions specified in paragraph 3.3.10 to accomplish three payload selectivity functions when used with the control unit and programmer unit described in paragraph 3.5.2 and Appendix A. The sequencer switch shall be functionally subdivided into two independent sections (10 and 20). Each section of the sequencer switch will advance in single steps when supplied with a voltage pulse (ground-seeking, 85 ±5 ms) from the programmer. The sequencer switch shall provide to the squib a minimum of 5 amperes, from a source other than the programmer to insure the "all-fire" requirements (5 amperes, 50 ms, minimum duration) of the MK 131, Mod 0 squib.

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3.5.3.1.1 (U) Safety.- The sequencer switches and programmer combination for the AN/ALE-39 dispenser system will be designed such that all squibs in the dispenser will be automatically shorted without the necessity of electric power. During payload dispensing, all squibs except the one selected for firing shall remain shorted. In addition, shorting of a squib shall recur automatically upon selection of subsequent squibs for firing to safeguard any misfired squibs. The sequencer switches shall be provided with an electrical interlock such that no power may be applied to the squibs unless the AN/ALE-39 programmer unit is connected in the wiring harness.

3.5.3.1.2 (U) Power loss.- In the case of a power loss the sequencer switch shall retain the position of the next squib to be fired in the dispensers.

3.5.3.2 (U) Form factor.- The form factor of the sequencer switch shall be 4.0 by 1.8 by 6.5 inches, including connectors, and shall be physically interchangeable with the AN/ALE-29A Sequencer Switch mounting provisions. Reference Figure 3.

3.5.3.3 (U) Weight.- The weight of each sequencer switch shall not exceed 3.5 pounds.

3.5.3.4 (U) Contents.- The sequencer switch shall contain circuits and such other components required to accomplish the functions listed above.

3.5.3.5 (U) Controls.- Not applicable.

3.5.3.6 (U) Connectors.- The connector selected for the interface with the dispenser module shall be MS 3113H-18-32P (or equivalent) and for the interface with the wiring harness shall be MS 3113H-12-10PX (or equivalent).

3.5.4 (U) Dispenser modules.- Characteristics of dispenser modules to be used with the AN/ALE-39 system are shown in Appendix A.

3.5.5 (U) Dispenser housings.- Characteristics of the dispenser housings to be used with the AN/ALE-39 System are described in Appendix A.

3.5.5.1 (U) Connectors.- The connector providing interface with the sequencer switch shall mate with a MS 3113H-18-32P.

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4. (U) QUALITY ASSURANCE PROVISIONS

4.1 (u) Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 (U) Classification of tests.- Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements:

(1) Preproduction tests

(2) Initial production tests

(3) Acceptance tests

(4) Life tests

4.2 (U) Preproduction (first article) tests.- Preproduction tests shall be made on an equipment representative of the production equipment to be supplied under the contract. Preproduction tests shall be accomplished under the approved test procedure of 4.6. The government inspector and the procuring activity shall be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory acceptable to the Government.

4.2.1 (U) Preproduction (first article) test data.- The contractor shall submit all data collected in conducting these tests to the procuring agency for review.

4.2.2 (U) Scope of tests.- Preproduction tests shall include all tests deemed necessary by the procuring activity to determine that the equipment meets all the requirements of this specification and the contract. Preproduction tests shall include environmental tests in accordance with the procedures of specification MIL-T-5422.

4.2.3 (U) Preproduction (first article) approval.- Approval of the preproduction sample shall be by the procuring activity upon satisfactory completion of all tests. No production equipment shall be delivered prior to the approval of the preproduction sample. Prefabrication of production equipment prior to the approval of the preproduction sample is at the contractor's own risk. The approved preproduction sample shall be retained by

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the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The preproduction sample(s) shall not be considered as one of the equipments under the contract.

4.2.4 (U) Production equipments.- Equipments supplied under the contract shall in all respects, including design, construction, workmanship, performance, and quality, be equivalent to the approved preproduction sample. Each equipment shall be capable of successfully passing the same tests as imposed on the preproduction sample. Evidence of non-compliance with the above shall constitute cause for rejection for equipment already accepted by the Government. It shall be the obligation of the contractor to make necessary corrections as approved by the procuring activity.

4.3 (U) Initial production tests.- One of the first ten production equipments shall be selected and sent at the contractor's expense to a designated Government laboratory for tests. This equipment shall be selected by the procuring activity after the equipment has successfully passed all individual tests. No other tests shall be conducted on equipment prior to starting the initial production tests. The preproduction sample shall not be selected for this test.

4.3.1 (U) Scope of tests.- This equipment may be subjected to any and all tests the procuring activity deems necessary to assure that the production equipment is equivalent to the previously approved preproduction sample in design, construction, workmanship, performance, and quality and that it meets all applicable requirements.

4.3.2 (U) Accessory material.- In addition to the complete equipment submitted for initial production tests the contractor shall also submit such necessary material and data necessary to test the equipment.

4.3.3 (U) Initial production sample approval.- Approval of the initial production sample shall be by the procuring activity upon satisfactory completion of all tests. Any design, material, or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Failure of the initial production sample to pass any of the tests shall be cause for deliveries of equipment under the contract to cease until proper corrective action is approved and accomplished. Corrective action shall also be accomplished on equipment previously accepted when requested by the procuring activity.

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4.3.4 (U) Reconditioning of initial production test sample.- On completion of the initial production test the equipment shall be reworked by the contractor by replacing all limited life or damaged items. After reworking, the contractor shall resubmit the equipment for acceptance.

4.4 (U) Acceptance tests.- The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the government inspector. Contractors not having adequate facilities for conducting all required tests shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall furnish test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

(1) Individual tests

(2) Sampling tests

(3) Reliability assurance tests

(4) Special tests

4.4.1 (U) Individual tests.- Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests:

(1) Examination of product

(2) Operational test

(3) Manufacturing run in test

4.4.1.1 (U) Examination of product.- Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.

4.4.1.2 (U) Operational tests.- Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation.

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4.4.1.3 (U) Manufacturing run in test.- Each equipment shall be operated under the conditions specified herein for a period of ten hours without failure. A failure shall be defined as anything which causes malfunctioning of the equipment. Only those adjustments will be permitted which can be made by using such controls and adjustment that are accessible to the operator during the normal use of the equipment. This test shall be deleted if the reliability test includes a test on each equipment which consumes at least ten hours of operation.

Temperature:	Ambient room
Humidity:	Ambient room
Vibration:	Any selected frequency within the range of 20 to 30 cps (excluding resonant points) and a minimum ampilitude of ±3 g's.

The equipment shall be vibrated (without vibration isolators) for a period of 10 minutes prior to the beginning of the 10 hour period of operation. Where feasible, the equipment shall be operated during this vibration period for the purpose of detecting flaws and imperfect workmanship. Operation within the specified limits of satisfactory performance is not necessarily required during the vibration period. The direction of vibration should be vertical to the normal mounting plane for 5 minutes and lateral to the plane for 5 minutes. Where it is not feasible to vibrate the equipment in 2 directions the vertical direction shall be used. During the 10-hour period of operation following the 10-minute vibration period, the equipment shall be mechanically cycled periodically through its various phases of operation. Should a failure occur, it should be repaired and the test started over, except that the 10-minute vibration period need not be repeated when it is certain the failure was not a result of the vibration. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future equipment. A record shall be kept of all failures. The 10-hour period specified above may be composed of two 5-hour periods to conform with standard working hours.

4.4.2 (U) Sampling tests.- Not required.

4.4.3 (U) Reliability assurance tests.- Reliability assurance tests shall be conducted using MIL-STD-781. Tests as required by both the Qualification Phase and the Production Acceptance (Sampling) Phase shall be conducted. Classification of failure shall be in accordance with MIL-STD-781 and AR-34.

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4.4.3.1 (U) Qualification phase.- Prior to the acceptance of equipments under the contract or order, minimum of three (3) equipments shall be tested as outlined in MIL-STD-781, under the section entitled "Qualification Phase of Production Reliability Tests." The maximum number of equipments to be used shall be those listed in Table 5 of MIL-STD-781. For the Qualification Phase, Test Level G shall be used. The Accept-Reject Criteria for Test Plan III shall be used.

4.4.3.2 (U) Reliability Production Acceptance Phase Tests.- The equipment, throughout production, shall be tested as outlined in MIL-STD-781 (as modified herein) under the section entitled "Production Acceptance (Sampling) Phase of Production Reliability Tests." Test Level G shall be used.

4.4.3.2.1 (U) All equipment test.- Each equipment produced except those submitted for the reliability qualification test, shall be tested for 50 hours. Prior to the 50-hour test on each equipment, a burn-in period is to be used at the option of the contractor. If the burn-in period is to be used the details thereof must be included in the approved test procedures. To determine whether the MTBF is being met at any time during the contract the operating test hours and the failures thereon (not counting burn-in failures or burn-in operating time) shall be totaled and the results compared with the reject line of Test Plan II of MIL-STD-781. (Extend the line as necessary to accomodate the data.) These totals shall accumulate so that at any one time the experience from the beginning of the contract is included. At the conclusion of each month the test results shall be sent to the procuring activity and to the Naval Air Systems Command, Attention: Avionics Division. At any time that the current totals of test hours and test failures plotted on Test Plan II curves show a reject situation, the procuring activity shall be notified. The procuring activity reserves the right to stop the acceptance of equipment at any time that a reject situation exists pending a review of the contractor's efforts to improve the equipment, the equipment parts, the equipment workmanship, etc., so that the entire compilation will show other than a reject decision.

4.4.3.3 (U) Test details.- The test details in addition to those included herein shall be part of the test procedures to be submitted and approved by the procuring activity prior to the beginning of the Qualification Test Phase of the Reliability Assurance Tests.

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4.4.4 (U) Special tests.- Special tests shall be conducted on a quantity of equipments for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the sampling or reliability assurance tests.

4.4.4.1 (U) Special test schedule.- Selection of equipment for special tests shall be made as follows:

(1) On an early equipment after an engineering or material change.

(2) Whenever failure reports or other information indicate additional tests are required. (This will be determined by the procuring activity.)

4.4.4.2 (U) Scope of tests.- Special tests shall consist of such tests as approved by the procuring activity. Test procedures previously approved for the preproduction tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests.

4.4.5 (U) Equipment failure.- Should a failure occur during either the sampling, reliability assurance, or special tests, the following action shall be taken:

(1) Determine the cause of failure.

(2) Determine if the failure is an isolated case or design defect.

(3) Submit to the procuring activity for approval, proposed corrective action intended to reduce the possibility of the same failure(s) occurring in future tests.

(4) Where practical, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

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4.5 (U) Life test.- The contractor shall furnish all samples and shall be responsible for accomplishing the life test. The test shall be of 300-hours duration and shall be conducted on equipment that has passed the individual test. The life test shall be performed under the conditions specified in 4.5.1. The life test sample shall be selected by the government inspector in accordance with the following: (Equipment which have successfully passed the Initial Production test, Sampling tests, Reliability tests, or Special tests may be selected for life tests. Test Life accumulated during the Reliability Test may be counted toward the Life test provided the entire 300 hours is accumulated on a single equipment and equipment selected is in accordance with the table below.)

Quantity of Equipments Offered for Acceptance	Quantity to be Selected For Life Test
First 25	1
Next 175	1
Next 300	1
	1 for each additional 500 or fraction thereof

4.5.1 (U) Test conditions.- The life test shall be conducted under the following simulated service conditions:

Temperature: Normal

Altitude: Normal ground (0-5000 Ft.)

Humidity: Room ambient

D. C. Voltate: 27.5 $\pm$2.0 volts

4.5.2 (U) Test periods.- The test may be run continuously or intermittently. Any period of operation shall be of sufficient duration to permit the equipment temperature to stabilize. Periodically, the equipment shall be turned on and off several times and put through its various phases of operation.

4.5.3 (U) Performance check.- At approximately 8-hour intervals during the test, a limited performance check shall be made. The performance check proposed by the contractor shall be subject to approval by the procuring activity.

4.5.4 (U) Test data.- The contractor shall keep the daily record of the performance of the equipment, making particular note of any deficiencies or failures. In the event of part failures, the defective part shall be replaced and the operation resumed for the balance of the test period. A record shall be kept of all failures throughout the test, including all tube failures. This record shall indicate the following:

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(1) Part type number

(2) The circuit reference symbol number

(3) The part function

(4) Name of the manufacturer

(5) Nature of the failure

(6) The number of hours which the part operated prior to failure.

4.5.4.1 (U) Failure report.- In the event of a failure, the government inspector shall be notified immediately. A report shall be submitted to the procuring activity upon completion of the test. In this report, the contractor shall propose suitable and adequate design or material corrections for all failures which occurred. The procuring activity will review such proposals and determine whether they are acceptable.

4.6 (U) Test procedures.- The procedures used for conducting preproduction tests, acceptance tests and life tests shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity or the government inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. Specification MIL-T-18303 shall be used as a guide for preparation of test procedures. When approved test procedures are available from previous contracts such procedures will be provided and may be used when their use is approved by the procuring activity. However, the right is reserved by the procuring activity to require modification of such procedures, including additional tests, when deemed necessary.

4.7 (U) Reconditioning of test equipment.- Equipment which has been subjected to acceptance and life tests shall be reconditioned by the contractor by replacing all wear or damaged items. After reworking the contractor shall resubmit the equipment for acceptance.

4.8 (U) Presubmission testing.- No item, part, or complete equipment shall be submitted by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

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4.9 (U) Rejection and retest.- Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the government inspector.

5. (U) PREPARATION FOR DELIVERY

5.1 (U) General.- All major units and parts of the equipment shall be preserved, packaged, packed, and marked for the level of shipment specified in the contract in accordance with specification MIL-E-17555 and MIL-STD-794.

6. (U) NOTES

6.1 (C) Intended use.- The equipment is intended to be used to dispense countermeasure decoys from military aircraft. These decoys are intended to be effective against weapon control radars, semi-active missiles and infra-red seekers.

6.2 (U) Ordering data.- Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(1) Title, number, and date of this specification.

(2) Selection of applicable levels of packaging and packing (see 5.1).

6.3 (U) Precedence of documents.- When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract: The contract shall have precedence over any specification.

(2) This specification: This specification shall have precedence over all applicable subsidary specifications. Any deviation from this specification or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

(3) Referenced specifications: Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

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6.4 (U) Performance objectives.- Minimum size and weight simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts, and processes other than those specified in specification MIL-E05400, it is desired their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accompanied by complete supporting information.

6.4.1 (U) Non-repairable subassemblies.- As a general rule non-repairable subassemblies should be encapsulated or hermetically sealed. The number of connections internal to the subassembly should be held to a minimum. Detail parts tolerances and ratings should be so selected that the life of the subassembly is greater than that of a similar repairable one. With few exceptions (such as high voltage power supplies), the non-repairable subassembly should evidence a Mean-Time-To-Failure greater than 5000 hours, and for many applications this figure must be nearer 50,000 hours.

6.5 (U) Type designations.- Parentheses () when used in the type designation, will be deleted or replaced by either a number or letter furnished by the procuring activity upon application by the contractor for assignment of nomenclature in accordance with 3.3.8. The complete type number shall be used on nameplates, shipping records, and instruction books, as applicable.

6.6 (U) Revisions.- In specification revisions and superseding amendments an asterisk "*" preceding a paragraph number denotes paragraphs in which changes have been made from the previous issue. This has been done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written, irrespective of the asterisk notations and relationship to the last previous issue.

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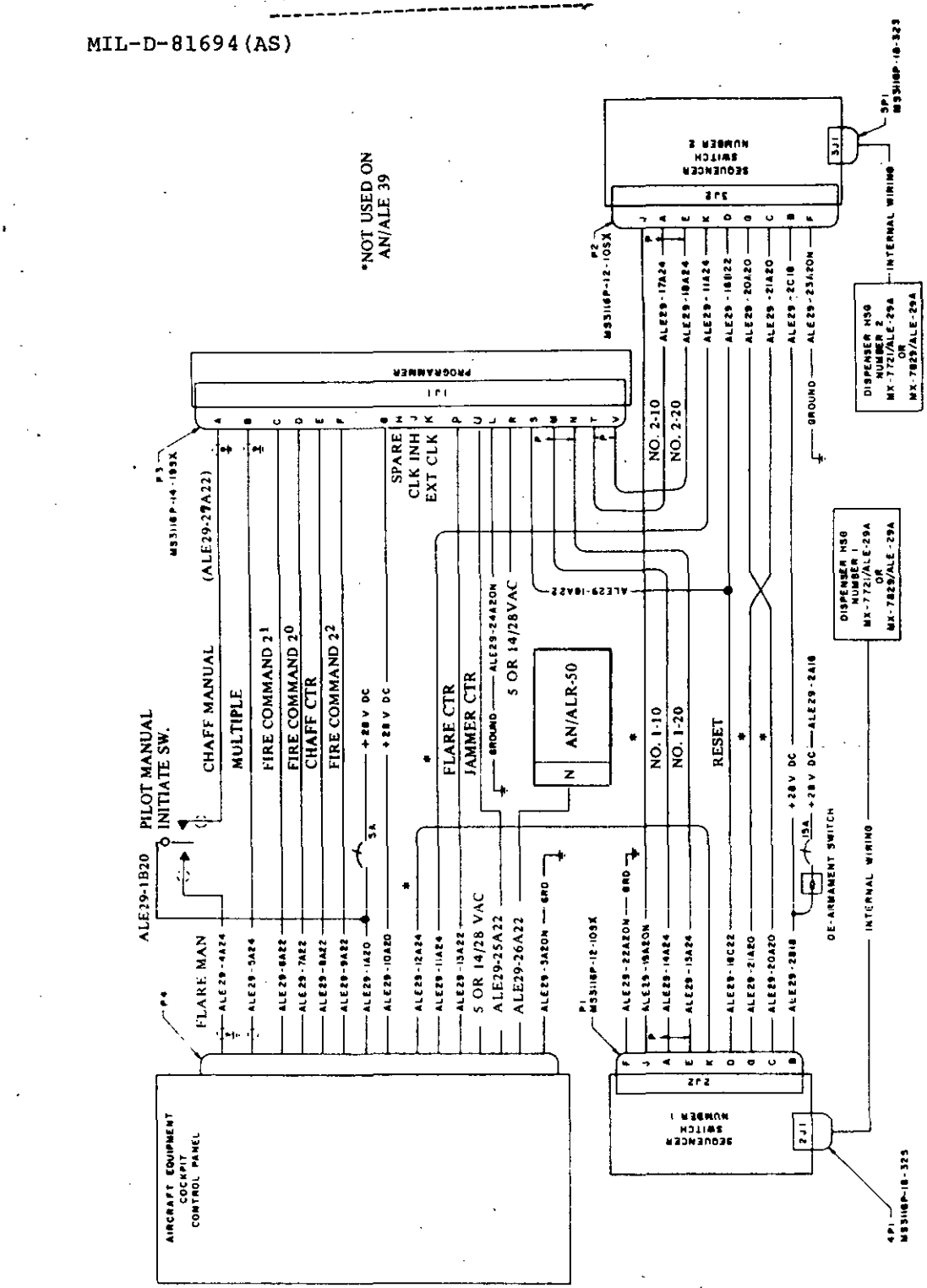


Figure 1 - AN/ALE-39 Countermeasures Dispensing Set

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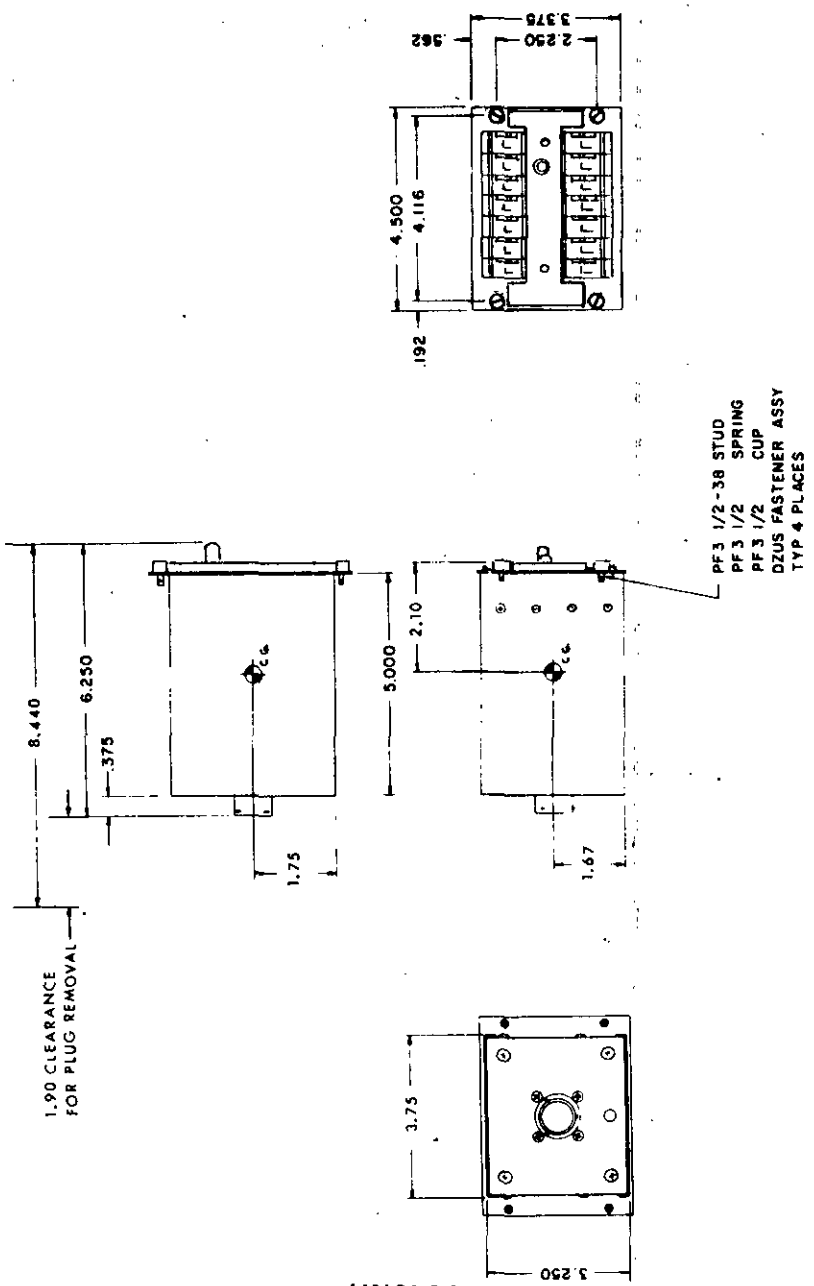


Figure 2 - Programmer

NOTES:

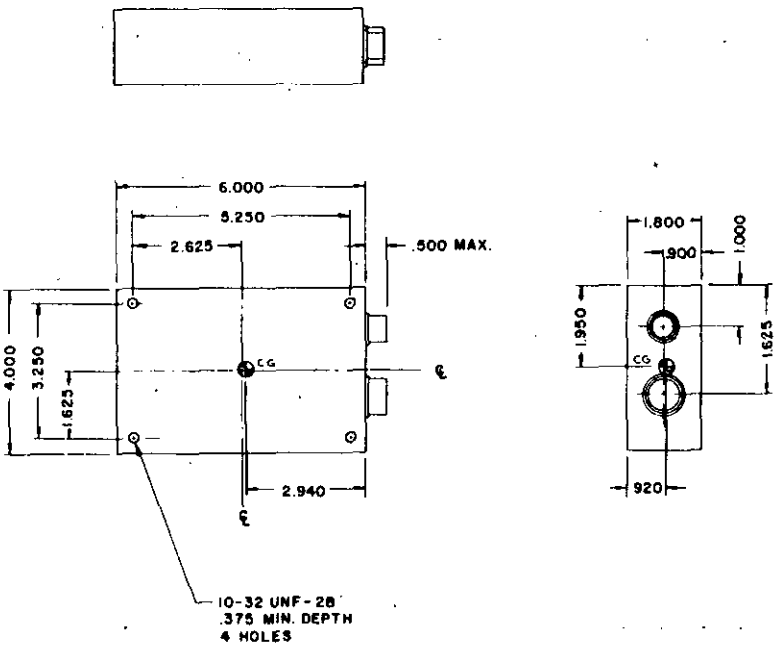
1. WEIGHT:
PROGRAMMER 1.9 LBS
2. HEAT DISSIPATION:
0.9 WATTS MAXIMUM

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2. HEAT DISSIPATION:
406 WATTS PEAK; 70% MAXIMUM DUTY CYCLE
1. WEIGHT:
SEQUENCER SWITCH 2.85 LBS.
NOTES:

Figure 3 - Sequencer Switch

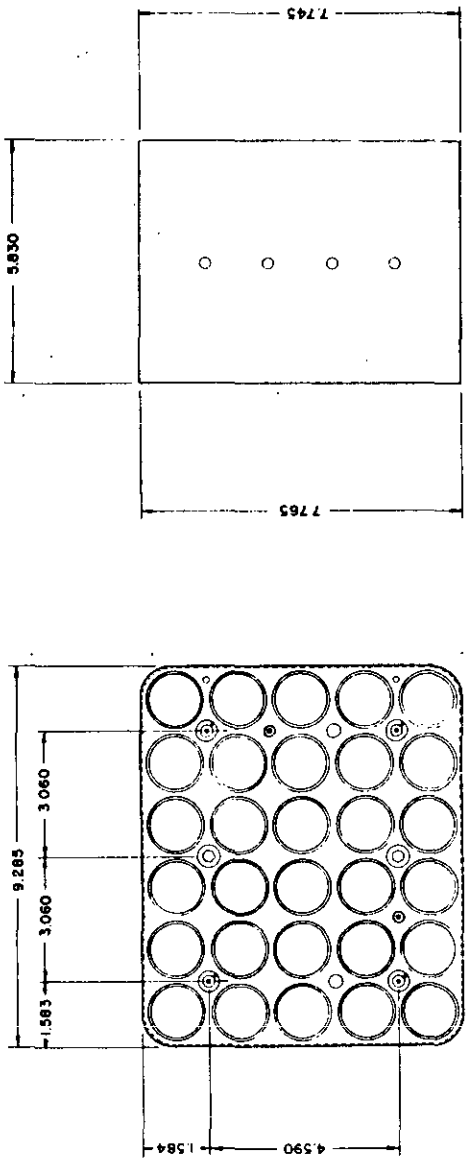
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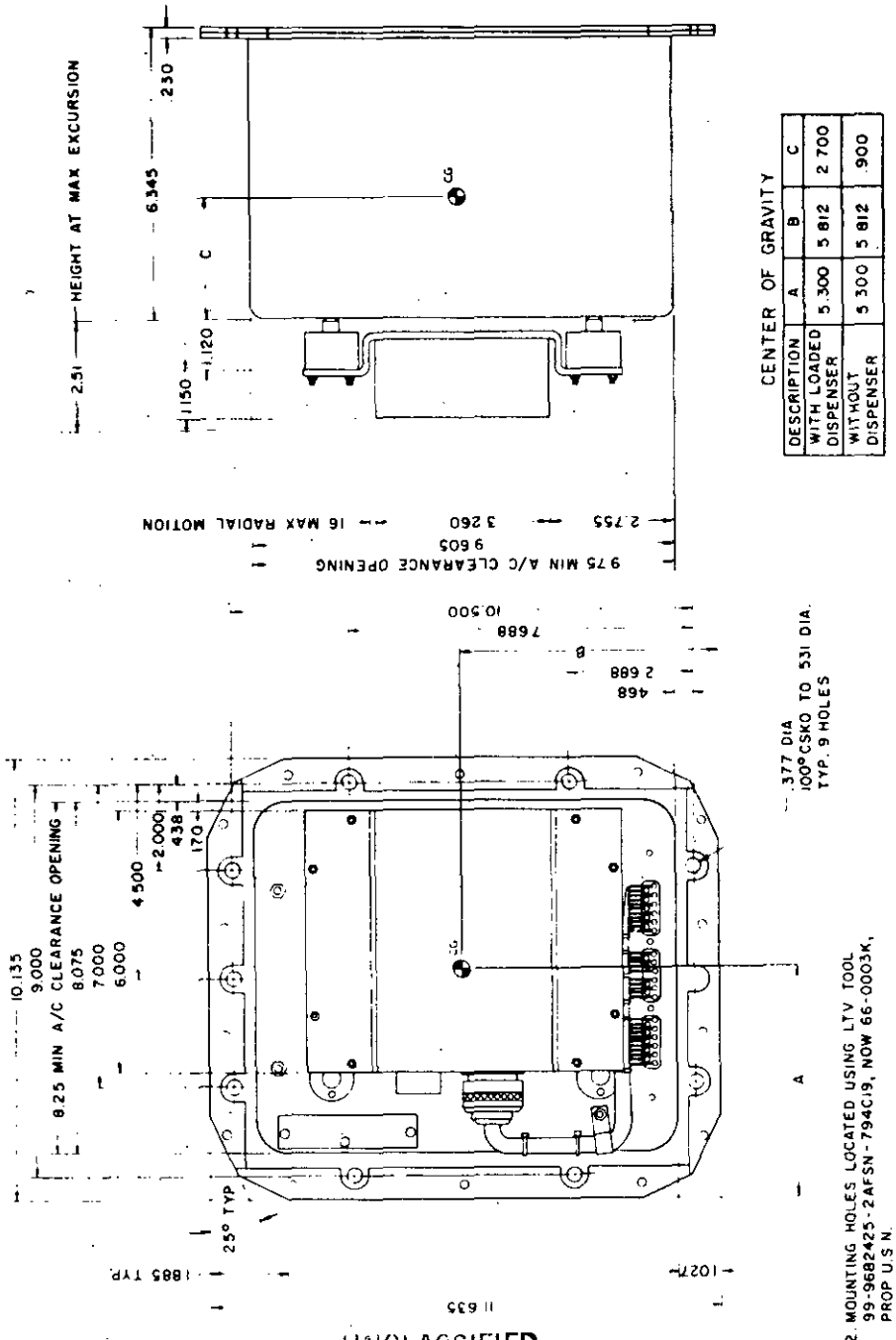


NOTES:
1. WEIGHT:
DISPENSER LOADED 21.30 LBS.
DISPENSER UNLOADED 8.20 LBS.
2. CENTER OF GRAVITY NOT APPLICABLE

Figure 4 - D-27/ALE-29A, Dispenser, Countermeasures

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CENTER OF GRAVITY			
DESCRIPTION	A	B	C
WITH LOADED DISPENSER	5.300	5.812	2.700
WITHOUT DISPENSER	5.300	5.812	.900

2. MOUNTING HOLES LOCATED USING LTV TOOL 99-9682425-2AFSN-794C19, NOW 66-0003K, PROP U.S.N.

1. WEIGHT
HOUSING - 4.66 LBS
SEQUENCER SWITCH WITH MOUNT - 3.70 LBS
DISPENSER (LOADED) - 21.30 LBS

NOTES

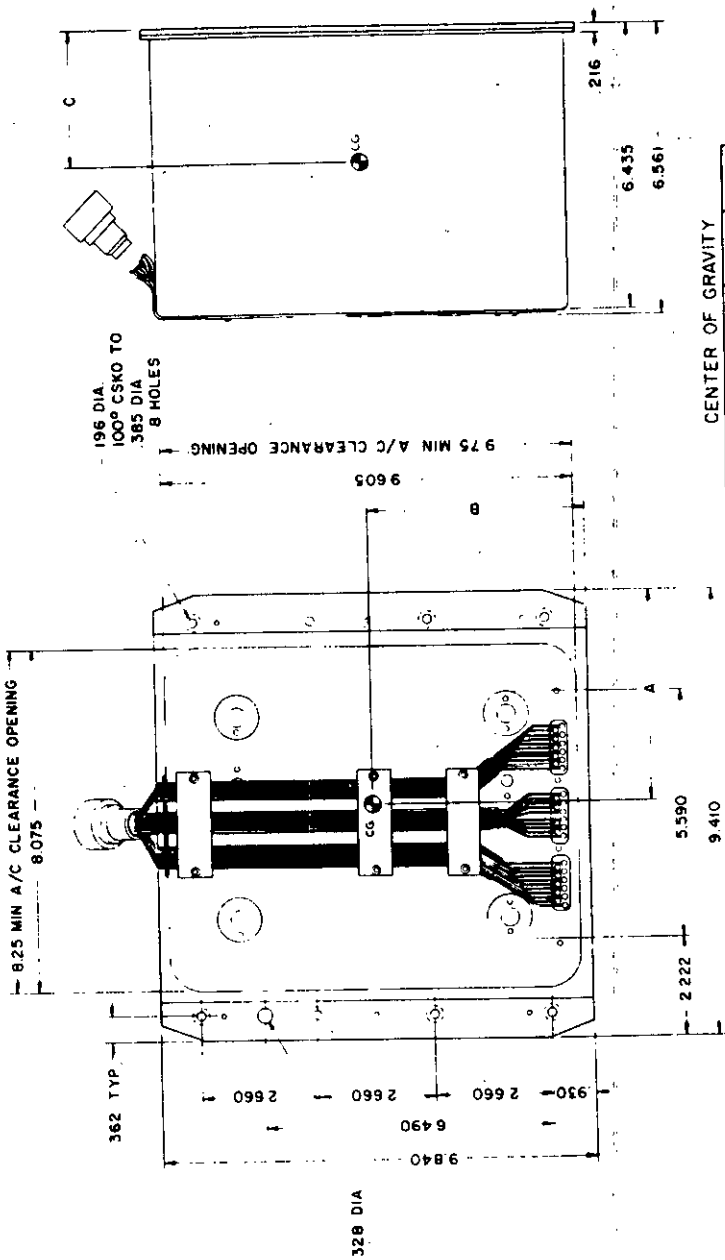
Figure 5 - MX-7829/ALE-29A Dispenser Housing

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CENTER OF GRAVITY

DESCRIPTION	A	B	C
WITH LOADED DISPENSER	4.950	4.900	3.000
WITHOUT DISPENSER	4.950	4.900	3.900

NOTES:

1. WEIGHT
   HOUSING - 4.43 LBS
   DISPENSER LOADED - 21.30 LBS.

Figure 6 - MX-7721/ALE-29A Dispenser Housing

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6.7 (U) Associated equipment.-

Type Designation	Quantity	Applicable Paragraph
Aircraft Wire Harness Set	1	3.3.13.1
Control-Unit (1)	1	Appendix A
Dispenser Module D-27/ALE-29A	2	Appendix A
Dispenser Housing MX-7721/ALE-29A or MX-7829/ALE-29A	2	Appendix A

(1) The configuration, electric interface, etc. will vary between types and models of aircraft selected for the AN/ALE-39 Dispenser system and therefore, the control unit will be considered under separate specification(s).

6.8 (U) This specification is under the cognizance of AIR-53362F.

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APPENDIX A

10. (U) ASSOCIATED EQUIPMENT

10.1 (U) Scope.- The scope of this appendix is the definition of characteristics for the associated equipment listed in 6.7 other than the Aircraft Wire Harness Set, which is defined in 3.3.13.1.

10.1.1 (U) List of defined equipment.- The equipment defined is as follows:

Control unit	see 30.1
Dispenser modules	see 30.2
Dispenser housings	see 30.3

20. (U) APPLICABLE DOCUMENTS

20.1 (U) General.- Documents when referenced in this appendix are listed in 2.1.

30. (U) CHARACTERISTICS

30.1 (U) Control unit.-

30.1.1 (U) Function.- The control unit provides the primary interface between the operator and the dispenser system. The control unit also serves as the interface with the warning receiver, preflight test equipment and pilot manual inputs to generate the necessary command signals for the programmer.

30.1.2 (U) Form factor.- The form factor of the individual control panels is dependent on type and model of aircraft.

30.1.3 (U) Weight.- Minimum consistent with good design practice.

30.1.4 (U) Contents.- The control units shall have components and circuits required to perform the functions listed in 30.1.5.

30.1.5 (U) Controls.-

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30.1.5.1 (U) Power control.- A toggle switch shall be provided to control the primary power to the AN/ALE-39 dispenser system. The position shall be labeled ON and OFF.

"OFF" In this position, no primary power shall be provided to the programmer and control units.

"ON" In this position, primary power shall be provided to the control and programmer units of the AN/ALE-39 dispensing system.

NOTE: For aircraft with very little panel space, the power control function may be added to the payload selector control rotary switch.

30.1.5.2 (C) Payload Selector Control.- A single five-position rotary switch shall be provided for selection of type payload to be dispensed. The positions shall be labeled Chaff, Flare-single, Flare-multiple, Jammer, and Rapid Fire-Flare.

30.1.5.2.1 (U) Rapid-Fire/Flare Control.- A guarded push-to-turn position on the rotary payload selector control switch shall be provided to actuate the programmer flare-firing sequence (reference 3.5.2.5.5). The switch position shall be labeled Rapid Fire and shall have a striped background, as specified in MIL-C-81774, paragraph 3.5.5.6.

30.1.5.2.2 (U) Flare Single Control.- A position on the rotary payload selector control switch shall provide the capability to actuate the programmer single-flare manual or automatic dispensing sequence.

30.1.5.2.3 (U) Flare Multiple Control.- A position on the rotary payload selector control switch shall provide the capability to actuate the programmer multiple-flare manual or automatic dispensing sequence (reference 3.5. .5.4).

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30.1.5.2.3 (U) Mode Control.- A single, momentary, long-throw, positive action, three-position toggle switch shall be provided to operate in conjunction with the payload selector control to initiate the dispensing sequence. The switch shall be labeled MANUAL, STANDBY and AUTOMATIC.

"MANUAL" When the switch is moved to the "MAN" or lower position, a single burst of the respective payload shall be dispensed within 0.075 seconds. This switch position is to be momentary, and shall be ready for reinitiation upon release. The manual mode may also be initiated while an automatic dispensing sequency is in operation.

"AUTOMATIC" When the switch is moved to the AUTOMATIC position, the AUTOMATIC sequency command for the payload type selected (reference 30.1.5.2) shall be entered into the programmer. The programmer shall control the rates and quantities of the appropriate payload until the dispensing program is completed. This switch position is to be mementary, and will be ready for reinitiation upon release, provided the AUTOMATIC dispensing sequence has ended. After interruption of an AUTOMATIC sequence by a manually-initiated single burst, the AUTOMATIC sequence will then resume.

30.1.5.4 (U) Auxilliary-Chaff-Flare Control.- A single pole momentary switch shall be provided on the throttle, stick, or any similar easily-accessible area to initiate a manual flare or manual chaff function (reference Figure 1). Manual flare function shall be initiated by a +27.5 VDC signal from the CHAFF-FLARE switch to the control unit. Manual chaff function shall be initiated by a +27.5 VDC signal from the CHAFF-FLARE switch to the programmer unit (pin A) (reference 3.5.2.5.2).

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30.1.5.5 (U) Warning receiver override.- A toggle switch shall be provided to disconnect the interface circuit in the control unit from the automatic AN/ALR-50-warning receiver (J2-M) input. In the "AUTO" position the interface circuit shall create the CHAFF-AUTO fire command code when initiated by the warning receiver. A repeat delay circuit shall be utilized to prevent the CHAFF-AUTO code from being generated more often than 30 seconds apart.

30.1.5.6 (U) Circuit functions.- The circuits discussed shall be capable of interfacing with the programmer and ancillary equipments pertinent to the particular aircraft utilizing the aircraft wiring harness. The logic is positive-true with logical "0" defined as 0.0 to +1.0 volts, and logical "1" defined as +10.0 to +12.0 volts. In the logical "0" condition, the circuitry, shall be capable of "sinking" 3.2 ma. In the logical "1" condition the circuitry shall be capable of "sourcing" 0.3 microamperes.

30.1.5.6.1 (C) Fire command encoder.- Following table lists codes for each dispensing mode selected:

(C)

	Function	2^0	2^1	2^2	Multiple
1.	Chaff-Manual	0	1	1	X
2.	Chaff-Auto	1	0	1	X
3.	Flare Single-Manual	0	0	1	1
4.	Flare Single-Auto	1	1	0	1
5.	Flare Multiple-Manual	0	0	1	0
6.	Flare Multiple-Auto	1	1	0	0
7.	Jammer-Manual	1	0	0	X
8.	Jammer-Auto	0	0	0	X
9.	Rapid Fire-Flare	0	1	0	X
10.	(Rest state)	1	1	1	X

X - either state

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30.1.5.6.2 (U) Interface.- Circuits are required to have interface with other aircraft systems such as preflight test system and pilot's manual control to generate the basic fire command codes listed above. As such, the detailed requirements shall be treated under separate specification for the control units.

30.1.5.7 (C) Display counter.- A display counter to indicate the number of each type of payload remaining in the dispenser shall be provided. Each counter shall have a manual means of setting or resetting the display to all whole integers between 00 and 60. The counters shall have the mechanical stops at 00 and shall not be damaged by additional counter signals. The counters shall be 28 volt type capable of being operated with a 30 millisecond pulse with a 60 millisecond duty-cycle. The programmer unit will supply a ground during the active portion of the cycle and will be capable of "sinking" 200 milliamperes, maximum.

30.2 (C) Dispenser modules.- Two (2) D-27/ALE-29A dispenser modules shall be used in the AN/ALE-39 system and three (3) types of payloads will be accommodated between the two dispensers in a permutation of two arbitrary sections of 10 and 20 payloads in each dispenser.

30.2.1 (U) Function.- The dispenser shall make provisions for accepting and dispensing 30 cylindrical units having a maximum length of 5.80 inches and a diameter of 1.420 + .000 - .010 inch. The squibs (MK 131, Mod 0) shall have a dismeter of 0.625 ±.005 inch, a maximum length of 0.560 inch. The electrical connector pins shall be 0.040 ±.001 inch diameter and spaced 0.200 ±.003 inch apart. The dispenser shall also provide the necessary circuit connections from the sequencer switch to the squibs.

30.2.2 (U) Form factor.- The form factor for the dispenser modules shall be as shown in Figure 4.

30.2.3 (U) Weight.- The dispenser shall not exceed 10 pounds when empty of payloads.

30.2.4 (C) Contents.- The dispenser shall consist of 30 tubes, a circuit board, and such mounting provisions as required for installation in aircraft as specified with the approval of the procuring activity.

30.2.5 (U) Controls.- No controls are required.

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30.2.6 (U) Electrical connectors.- No connectors between the dispenser and the aircraft wiring are required since the interface is through the sequencer switch. The connector providing interface with the sequencer switch shall mate with a MS 3113H-18-32P (or equivalent).

30.3 (U) Dispenser housings.- The dispenser housings are of two basic types; the MX-7829/ALE-29A and the MX-7221/ALE-29A (reference Figures 5 and 6).

30.3.1 (U) Function.- Either of the two dispenser housings provides a mechanical interface between the airframe and the dispenser module. In addition, the MX-7829/ALE-29A also provides a mechanical mount for the sequencer switch. Also, the housing contains a printed wiring board for mounting of the squibs and the necessary connectors and wiring harness to interconnect the sequence switch and the squib board.

30.3.2 (U) Form factor.- The form factor of the MX-7829/ALE-29A and the MX-7221/ALE-29A dispenser housings shall be as shown in Figures 5 and 6

30.3.3 (U) Weight.- Weight of the dispenser housings shall not exceed 5.0 pounds.

30.3.4 (C) Contents.- The dispenser housings shall contain the provisions for mounting of the Dispenser Module D-27/ALE-29A and the necessary wiring harness for mating with the sequencer switch.

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